Topic Highlight Project

Autonomous mobile robot for follower task

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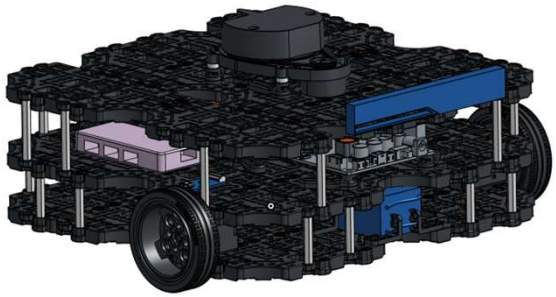
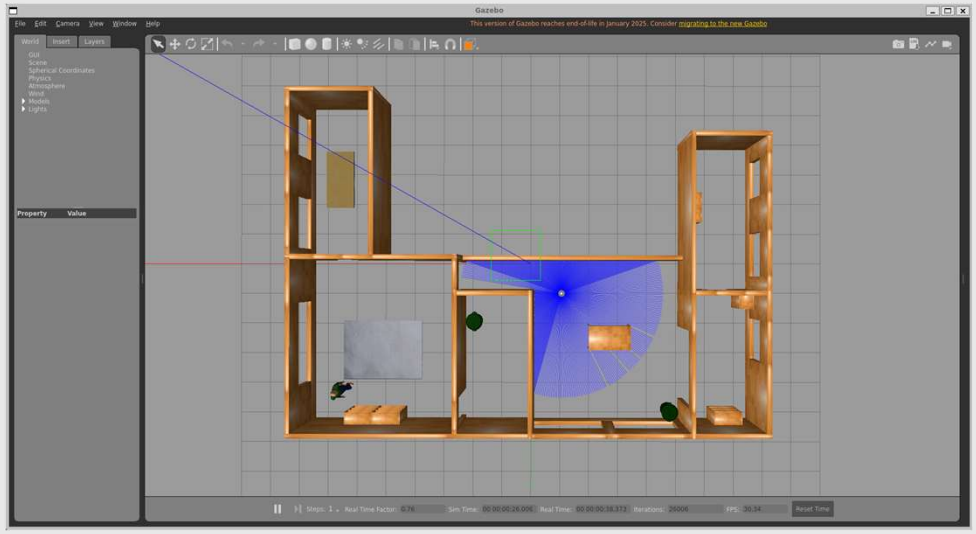
Objective

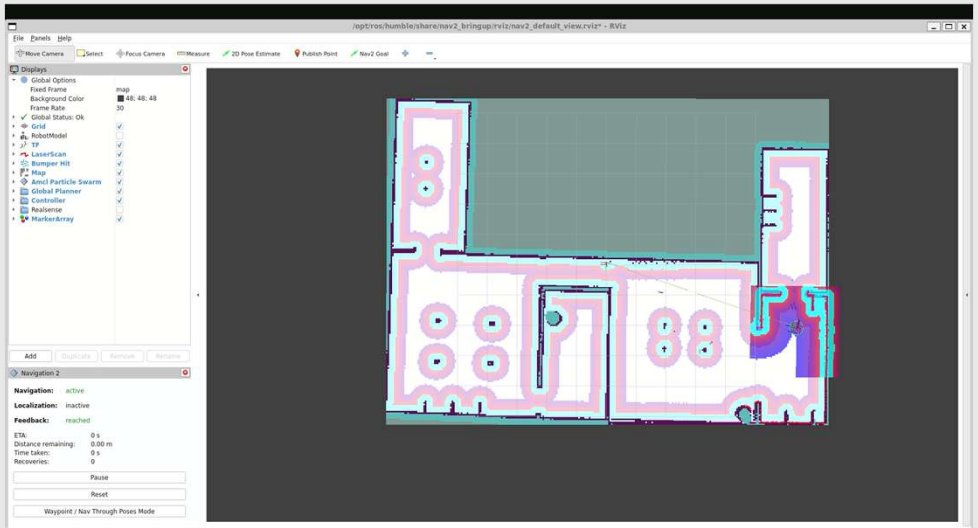
Using the TurtleBot3 Waffle with a RGB-D camera, develop an application that enables the robot to follow a designated target, identified by a Neural Network, while simultaneously mapping the environment as the target moves.



Simulation Environment

ROS 2™

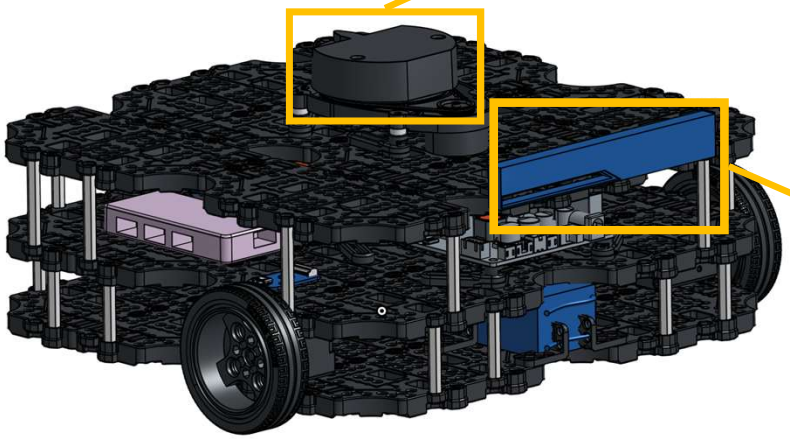




Simulation Tools

- ROS2 Humble
- Gazebo
- Rviz

Robot Description



LiDAR

Turtlebot3 waffle is equipped with a 360° LiDAR for SLAM (Simulataneous Localization And Mapping, technique that is used to estimate the current robot location in an arbitrary space) & Navigation purposes like obstacle avoidance.
The LiDAR driver reads raw scan data and publishes on the **/scan** topic (type: **sensor_msgs/LaserScan**).

Camera (Intel RealSense r200)

The camera is used to detect the target in the environmental space. The RealSense camera provides both RGB and depth data. Information are publishsed on **/camera/image_raw** and **/camera/depth/image_raw** respectively.

A Better Insight:

1) The frames coming form the RGB camera are analyzed in order to detect the moving target through an Identification Neural Network.

2) The Depht camera allows to acknowledge the distance from the robot.

Intel RealSense r200

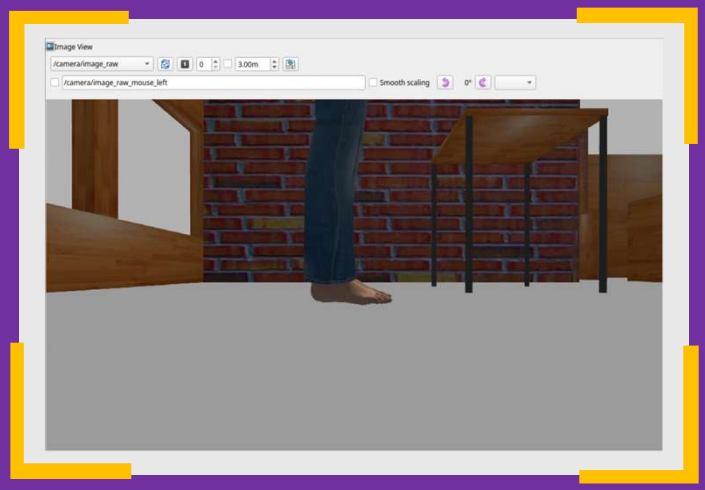
A RGB-D camera used for 3D vision, based on the pinhole camera model.
Design for robotics and computer vision application, provides RGB and depth datas.

1920 x 1080 @ 30 FPS
RGB resolution

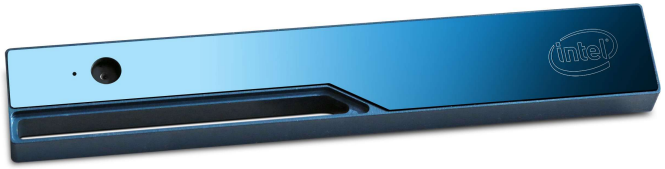
1920 x 1080 @ 60 FPS
Depth resolution

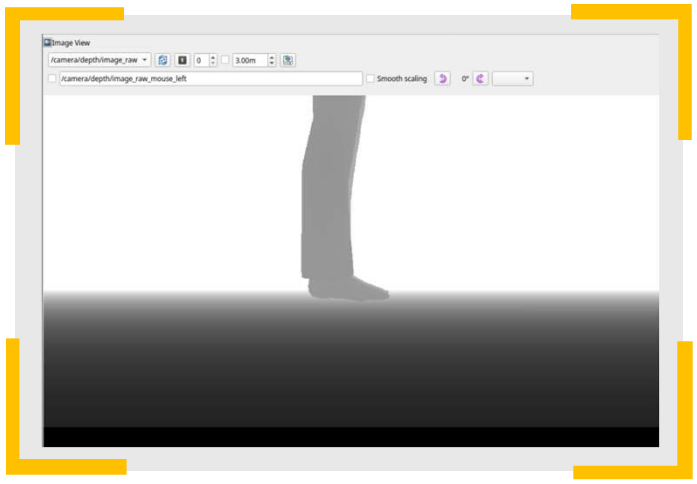
0.5 m to 3.5 m
Depth range

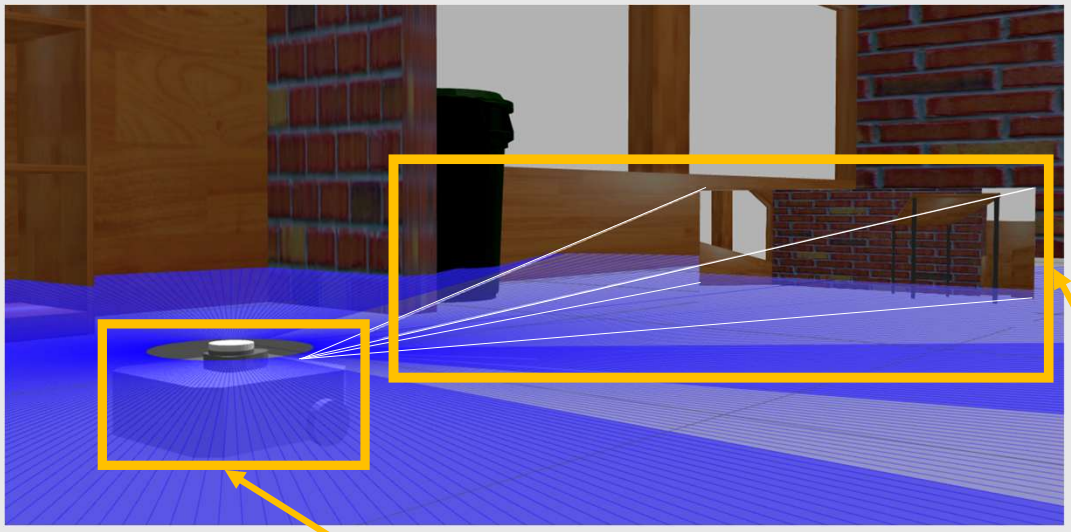
1.03 rad (≃ 59 °)
Horizontal Field Of View



Camera







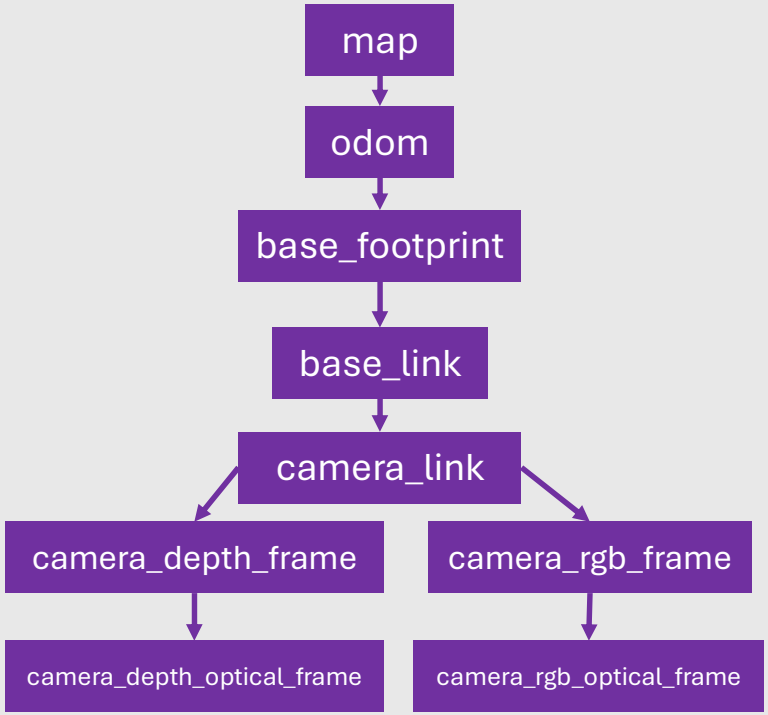
Turtlebot3 waffle

Two-wheeled unicycle model , the robot model is confined in the plane, the dynamic state comprises the plane coordinate and the twisting angle (yaw), its position and orientation are given with respect to the map frame.

F.O.V.

The field of view of the vision sensor is represented as a rectangular pyramid (illustrated with thin white lines). The position and orientation of this sensor are defined relative to the robot's **camera_link** frame, which in turn, is referenced to the **map frame** (inertial frame).

Camera Transformation Tree:



Target

Movable Target

In order to set up a movable and trackable target, a model of a walking person has been placed inside the chosen Gazebo world.

Person Model

The person model is detailed enough to allow the robot to detect individual body parts as well.

Trajectory Description

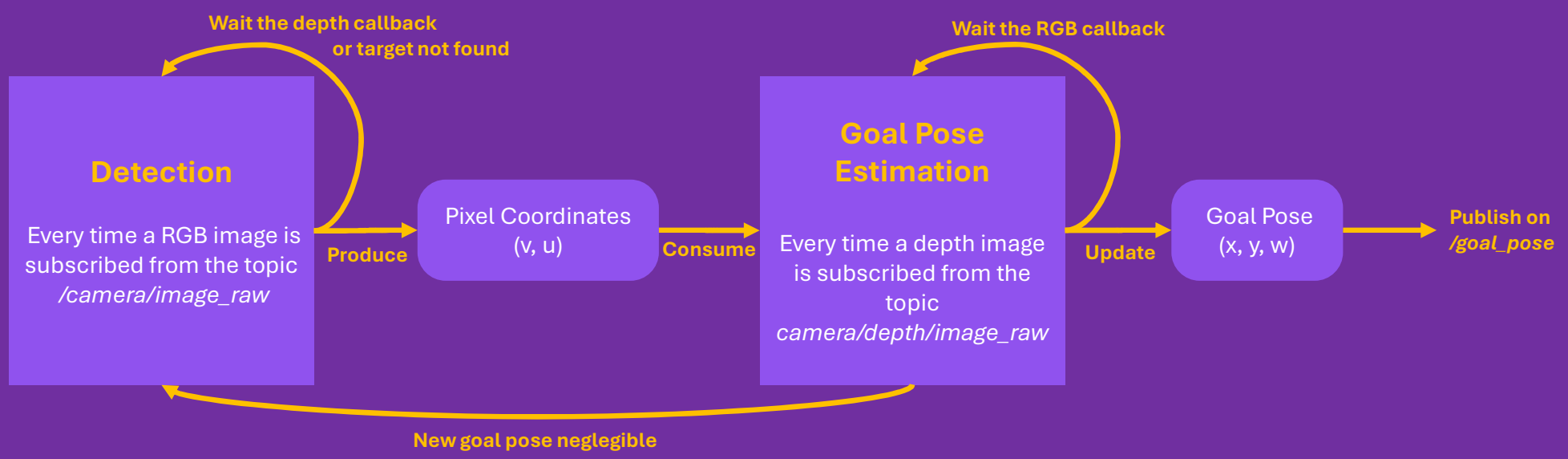
The person's motion has been defined so that he follows a reference trajectory described by several waypoints.

Trajectory Implementation

The trajectory was built experimentally according to the robot's horizontal F.O.V. In fact, we want to always frame the person.

Detection & Goal Pose Estimation

The implemented ROS2 node detetcts the person and publishes the coordinates of the **target pose** of the robot with respect to the **map frame** on the ***/goal_pose*** topic. This is a default topic of the Nav2 that automatically plans the optimal path. The processes have to be synchronized, in order to have only **one detection** active **at time**:



Detection Strategy

Detection Activate

Every time a RGB image is subscribed, while the depth callback is not active, the detection starts.

Person Detection

For the person detection we used a **Neural Network** in particular Mask R-CNN model pre-trained on COCO.

Frame Adaptation

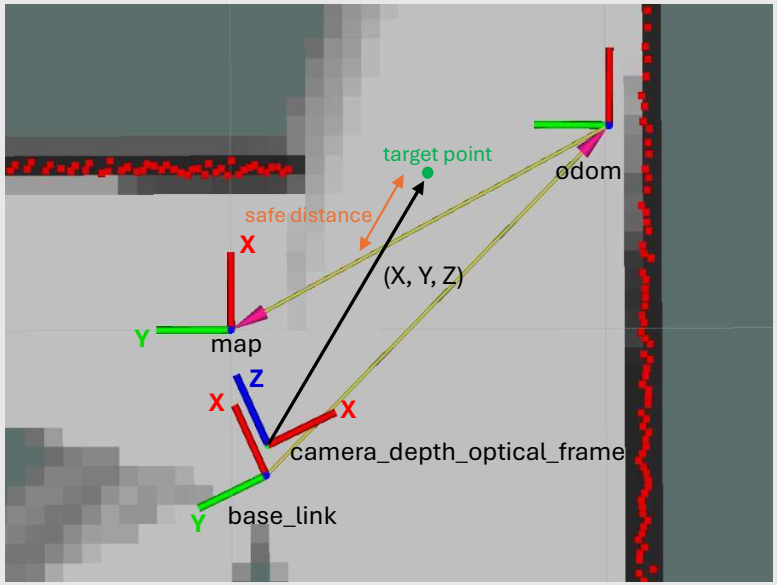
The detection is the most **time consuming** task, to speed up it, we reduced the size of the frames by a factor of 3: from (1920,1080) to (640,360).
The average detection time is 0.5 sec.

Target Point

As target point we have chosen the centre of the **bottom side of the bounding box** , in this way we avoid detection beyond the person.

Goal Pose Estimation

Every time a depth image is subscribed, while the previous detection is finished, the goal pose estimation starts.



Estimate the 3D coordinates of the point w.r.t. the **camera_depth_optical_frame**:

$$Z = image_depth[v, u]$$

$$X = \frac{u - c_x}{f_x} Z$$

$$Y = \frac{v - c_y}{f_y} Z$$

The intrinsic parameters of the camera are found in the topic **/camera/depth/camera_info**.
NOTE: since the RGB image is shrinked by a factor of 3, the pixel coordinates have to be multiplied by 3.

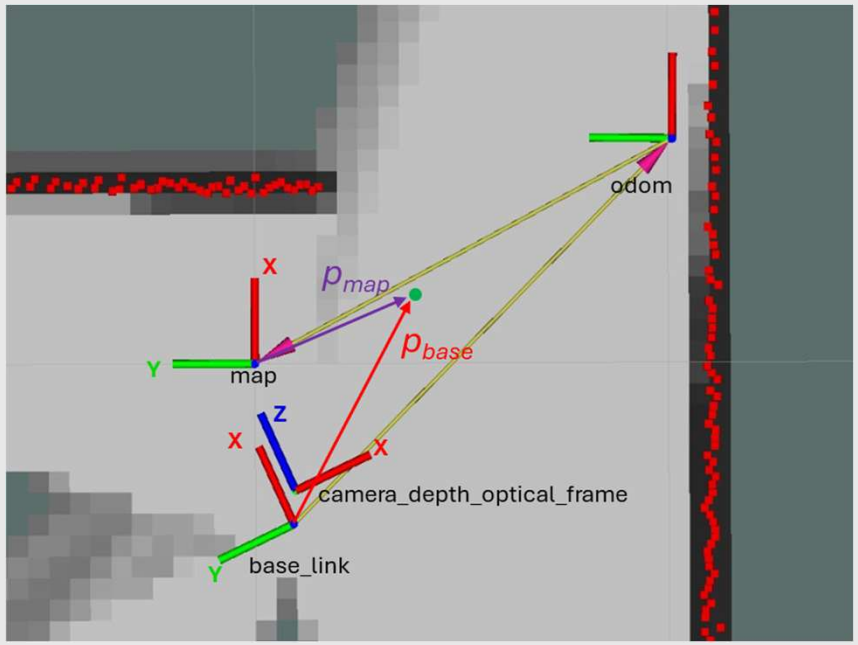
We have subtracted the **safe distance (0.6m)** to the camera coordinates, to avoid collision and to ensure that the person is properly framed.

Goal Pose Estimation

Starting from the coordinates in the camera frame, the goal **posistion** is computed in the map frame:

$$p_{map} = T_{map \to cam} \cdot p_{cam}$$

If the differance between the new goal position and the previous is less than 0.2m, the new goal is neglected.



To estimate the goal **orientation**, the base frame is rotated, in order to align its X-axis with the position vector.
The coordinates of the target point are computed w.r.t. the base frame:

$$p_{base} = T^{-1}_{map \to base} \cdot p_{map}$$

$$\theta = atan2(p_{base,y}, p_{base,x})$$

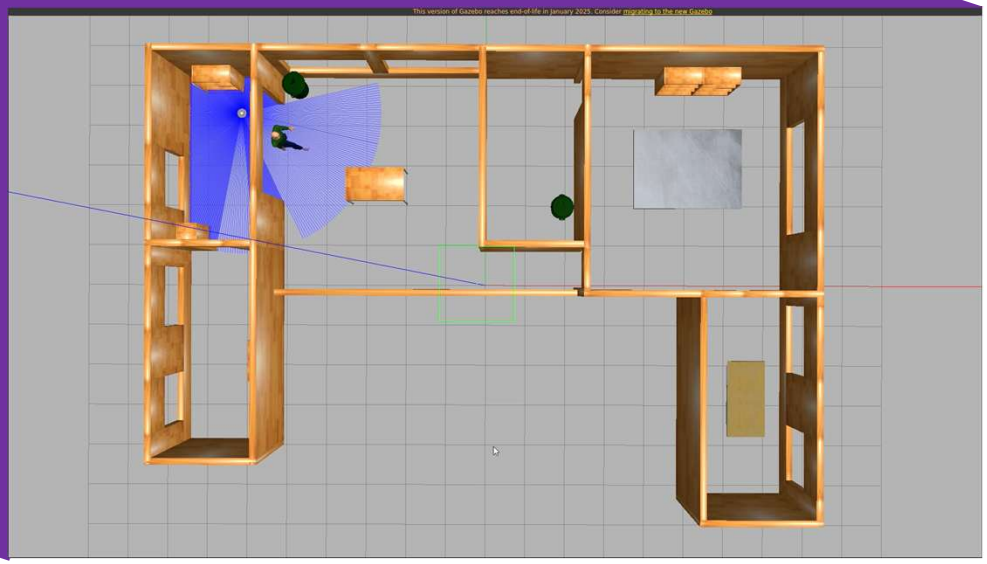
$$R = R_z(\theta) \cdot R_{map \to base}$$

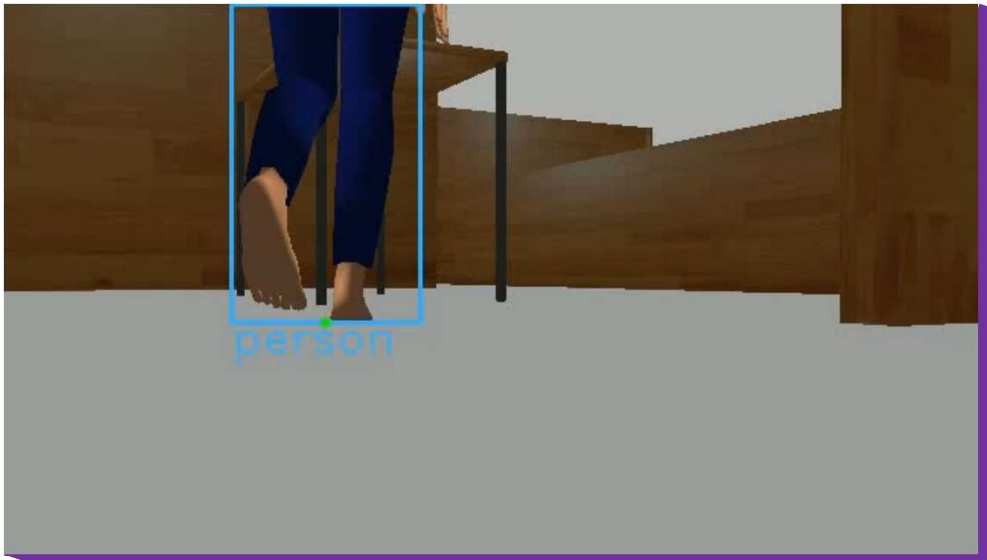
The Yaw angle (w) is estracted from R, that express the goal orientation of the robot w.r.t. the map frame.

The goal pose (x, y, w) of the robot w.r.t. the map frame is published on the **/goal_pose** topic, and **Nav2** with **slam-toolbox** takes care of the path planning.

First simulation

Gazebo



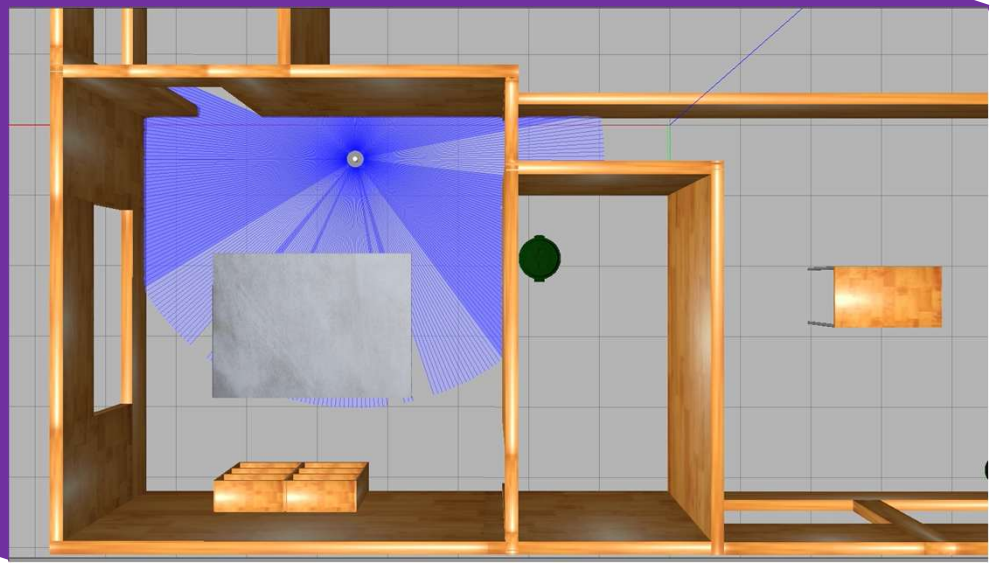


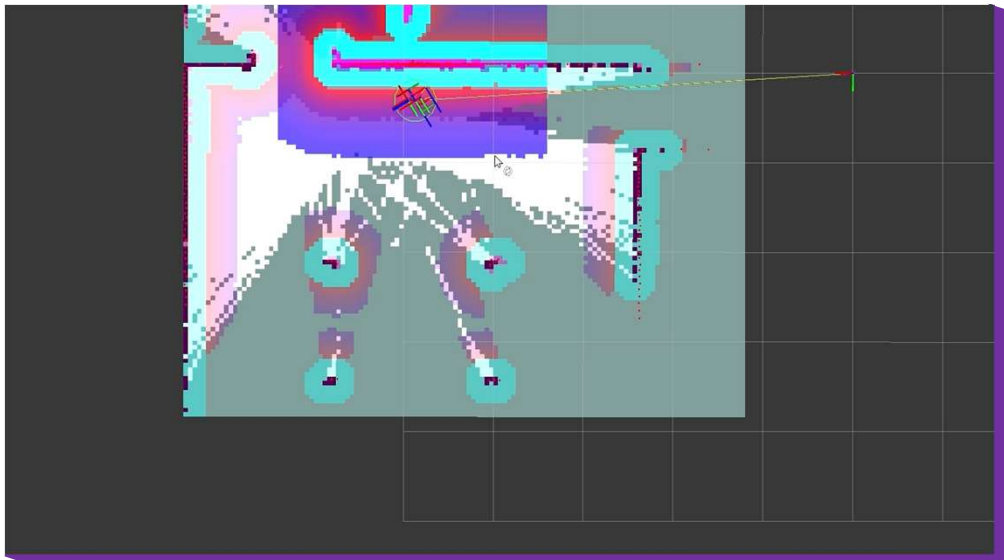
Bounding Box

The first simulation focuses on the TurtleBot's ability to follow the target at higher velocities while maintaining a safe distance.

Second simulation

Gazebo





RViz

The second simulation focuses on the TurtleBot's ability to follow the target in closed environments, even when obstacles are present between them.

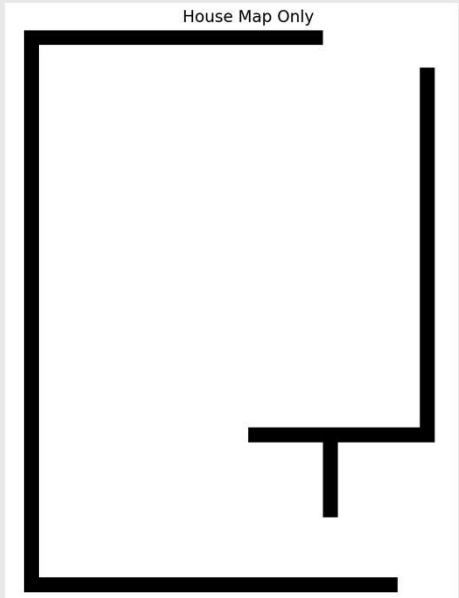
Problem

- It is assumed that the robot frames the person at the beginning of the simulation, but in general applications, this could be different;
- It is possible that the person performs a non-smooth trajectory, exiting the robot's FOV.

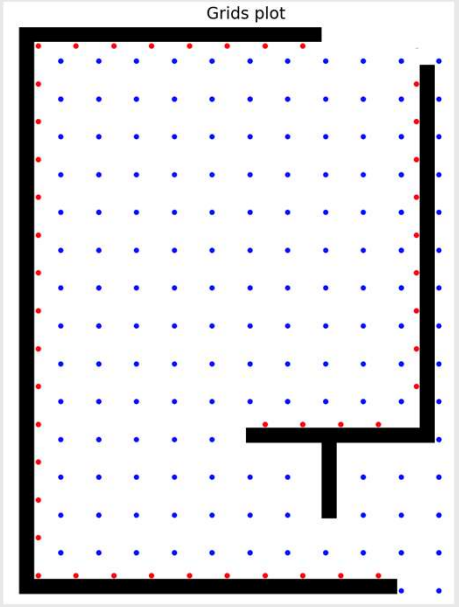
Solution

In order to help the TurtleBot3 detect the person, a grid coverage approach is used, where several monodirectional cameras are positioned to cover as large an area as possible.

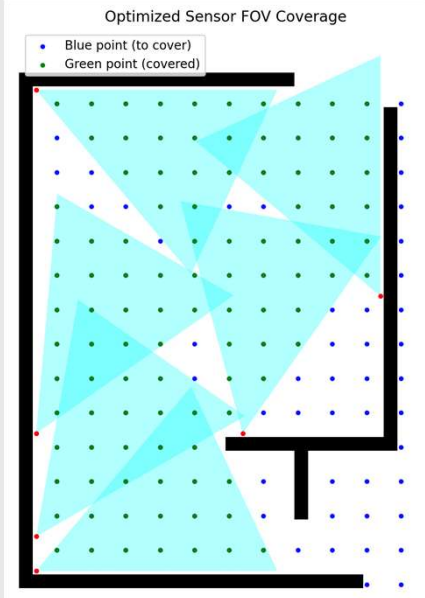
Grid coverage



A **2D representation** of a portion of the house environment has been designed for coverage purposes. As in the **grid coverage algorithm**, walls and free space are represented by boolean variables.



This algorithm uses two grids: a **sensor grid**, which represents the possible locations of the unidirectional sensor to be placed, and a **grid area** representing the area that must be maximized in terms of coverage by the FOV of the sensors.

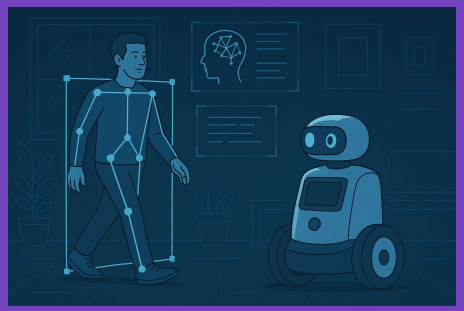


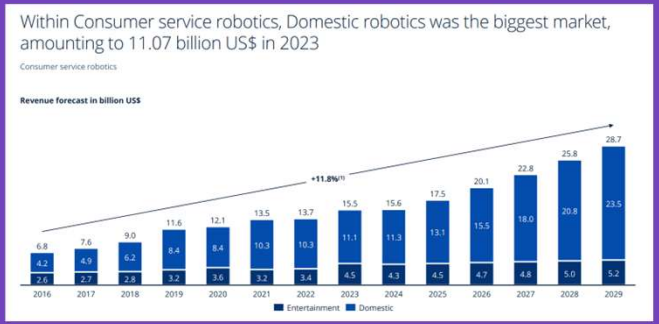
The **metaheuristic** algorithm employs a heuristic along with a selection criterion (maximize the range based on FOV and, for the FOV angle, the number of grid points detected) in order to connect to a global search of the **feasible decision space**. In the proposed image, the number of covered grid points for each sensor is 15.

Future Improvemens & Applications

Establish a **track coverage algorithm** integrated with the coverage algorithm to solve the problems when the robot loses the person.

Improve the robot's perception capabilities to understand the behavior of the person by also extracting **information from the environment**.



Possible applications of **assistive robots**: rehabilitation, companionship, mobility , and assisting a physically disabled user.

Growing interest in **Domestic robotics** in both research and industry.